

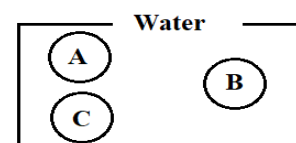
Engineering Physics II	Faculty of Engineering	Date: 3 /8/2025
B₁ A/E	First Test - Modeule C	
Name:	Reg. No.:	

Section I: Answer all the following by putting true (T) or false (F): **8 Marks**

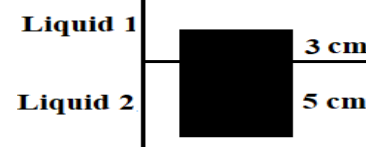
1) Archimedes principle related only to fully immersed object		
2) Electric flux depends on the charge inside		
3) Downward acceleration is for sinking object		

Section II: Answer all the following by selecting the correct choice: **14 Marks**

1) Less electric field lines directed inward				
a) - 1 C	b) 4 C	c) - 7 C	d) 9 C	e) None of these
2) If a ratio of fluid to object densities is 1. Then the fully immersed object is:				
a) Sinking	b) Floating	c) Moving up	d) At rest	e) None of these
3) 13 electrons and 6 neutrons. The total charge is:				
a) - 7 e	b) 7 e	c) - 13 e	d) 13 e	e) None of these
4) From the figure shown below. The pressures				



a) $P_C < P_A$	b) $P_A = P_C$	c) $P_B > P_C$	d) $P_B < P_C$	e) None of these
5) A cube of x each side is placed in $\vec{E} = \alpha \hat{i} + \beta \hat{j} - \gamma \hat{k}$. The electric flux in yz and xz planes is:				
a) $(\alpha + \beta) x^2$	b) $(\alpha - \gamma) x^2$	c) $(\beta - \alpha) x^2$	d) $(\beta - \gamma) x$	e) None of these
6) The gravitational force is given by:				
a) $G m/r^2$	b) $G m_1 m_2 / r^2$	c) $G m_1 m_2 / r$	d) $G m/r$	e) None of these
7) An object of ρ suspended between liquid 1 of ρ_1 and liquid 2 of ρ_2 . The object's density is:				



a) $\rho = \frac{3}{8} \rho_1 + \frac{5}{8} \rho_2$	b) $\rho = 5 \rho_1 + 3 \rho_2$	c) $\rho = \frac{5}{8} \rho_1 + \frac{3}{8} \rho_2$	d) $\rho = 3 \rho_1 + 5 \rho_2$	e) None of these
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Section III: Answer all the following by selecting the correct choice

Full and clear steps are required

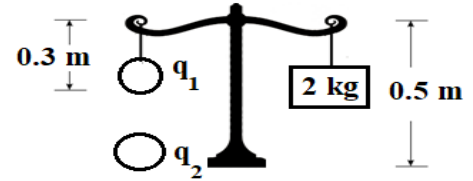
20 Marks

Constant		
Electron: $m = 9.11 \times 10^{-31}$ kg	$K_e = 9 \times 10^9$ N.m ² /C ²	Gravity: $g = 10$ m/s ²
Proton: $m = 1.67 \times 10^{-27}$ kg	$G = 6.67 \times 10^{-11}$ N.m ² /kg ²	Water: $\rho_f = 1000$ kg/m ³

1) 500 N object floats in water of $\rho_f = 1000$ kg/m³, the object is immersed by 8 cm and 11 cm over the water. The volume of the object in m³ is:

a) 0.086	b) 583.3	c) 0.119	d) 727.3	e) None of these
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2) 2 kg at one side of a balance scale and a charge $q_1 = 2 \text{ nC}$ at the other as shown. The value of the charge q_2 for equilibrium is:

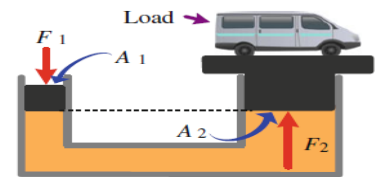


a) - 0.0444 C	b) 0.01778 C	c) 0.01481 C	d) - 0.0296 C	e) None of these
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3) $q_1 = 3 \text{ nC}$ is placed at $(-0.3 \text{ m}, 0)$, $q_2 = 3 \text{ nC}$ is placed at $(0, -0.3 \text{ m})$. The electric field in N/C at the origin is:

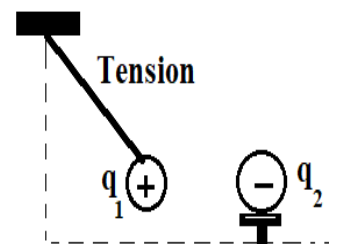
a) 282.8 at 315°	b) 282.8 at 135°	c) 424.3 at 135°	d) 424.3 at 45°	e) None of these
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4) In Pascal principle: The piston areas are 75 cm^2 and 250 cm^2 respectively, the weight on the output piston is 10000 N. The weight that the input piston handle is:



a) 3000 N	b) 33333.3 N	c) 333.3 N	d) 300000 N	e) None of these
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5) 100 g and $q_1 = 2 \text{ }\mu\text{C}$ charge hung by a rope. A charge of $q_2 = -6 \text{ }\mu\text{C}$ attracts q_1 charge. Then the rope inclined and the distance between the charges is 20 cm. The tension in the rope is:



a) 3.7 N	b) 4.2 N	c) 120.6 N	d) 2.9 N	e) None of these
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